Semiconductor FA Flow

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Technical summary | EFA / EFI / PFA | ATE and SLT correlation

Objective: Connect the electrical failure signature, physical defect and verified root cause. Select techniques by evidence; the sequence is iterative, not mandatory for every case.

01 Receive the case and define the failure

Collect: Device ID, lot, wafer coordinates, package type and revision; failure stage (wafer sort, final test, SLT, reliability stress or customer use); failing test, bin, logs, voltage, frequency and temperature.

Sample history: Rework, stress, handling and previous FA. Obtain a known-good comparison unit, preferably from a comparable lot/revision.

Initial suspects: Failure type (open, short, leakage, functional, timing, intermittent); location (board/package/die/region); block; trace/net; and pin/ball number, signal name and power domain. Mark suspects as unverified.

Output: Precise problem statement, sample record and initial suspect list.

02 Confirm and reproduce

Repeat the original failing test. Verify program revision, socket, fixture, board, cables and supplies. Swap DUTs/fixtures and compare with a good unit. Record repeatability, intermittency and environmental sensitivity.

Output: Confirmed DUT failure, marginal behavior, test-system issue or failure not reproduced.

03 Characterize the electrical signature - EFA

Signature	Checks
Open / high resistance	Continuity, resistance and pin I-V comparison
Short / leakage	Rail resistance, supply current and pin I-V
Functional / timing	Reduced sequence, logs/registers; voltage-frequency-temperature shmoo
Memory / scan	MBIST, fail-bit maps, pattern dependence; chain integrity, ATPG/scan diagnosis
Interface	Training logs, loopback, error counters and waveforms

Correlate wafer sort, final ATE, SLT and bench results. Use appropriate bias limits and current compliance. **Output:** Signature, failing conditions and suspected circuitry.

04 Rank hypotheses and plan tests

List plausible causes and supporting/conflicting evidence. Choose tests that distinguish board, package and silicon causes. Update suspected location, block, trace/net and pin as evidence develops.

Output: Ranked hypotheses and discriminating next-test plan.

05 Inspect the package nondestructively

Optical: Damage, corrosion, contamination, ball condition. **X-ray:** Interconnect/package abnormalities. **CSAM:** Delamination/interface abnormalities. **TDR:** Accessible interconnect discontinuities, where applicable. Correlate findings to the failing net; an image alone does not prove causality.

Output: Evidence supporting package involvement or further die investigation.

Localization to closure

Steps 06-10 | Preserve evidence and verify causality

06 Localize the fault - EFI

Apply the failure-reproducing bias or test sequence. Compare with good-unit behavior and map results to layout coordinates. Choose the method suited to the signature and available access.

Technique	Main application
PEM / emission microscopy	Abnormal emission from electrically active defects
Lock-in thermography	Localized abnormal power dissipation
OBIRCH / thermal laser methods	Resistive defects and conductive paths
LVP / LVI / optical probing	Dynamic logic and timing investigation
Scan / memory diagnosis	Candidate cells, nets and memory peripheral circuitry
Nanoprobing	Local transistor/interconnect characterization after access preparation

Output: Region of interest, candidate structures and localization confidence. A hotspot or diagnosis candidate requires confirmation.

07 Prepare the sample and recheck

Plan decapsulation, backside preparation or layer access. Preserve orientation, coordinates and connections; document preparation. Retest after significant steps where feasible. Investigate any change or loss of the original signature.

Output: Accessible sample with the failure preserved, or its change documented.

08 Perform targeted PFA

Use optical/SEM inspection, delayering and targeted FIB cross-sections. Add voltage contrast or nanoprobing for connectivity/electrical confirmation; EDS for detectable elemental composition; TEM/STEM for nanoscale structures. Check good structures and preparation artifacts.

Output: Physical findings correlated to the electrically implicated location.

09 Verify mechanism and classify root cause

Distinguish **symptom** (what failed), **mechanism** (how it failed) and **root cause** (why it occurred). Confirm agreement with electrical behavior, EFI location, good-unit comparisons and relevant history. Separate initiating defects from consequential damage.

Classification	Typical cause / status
Design issue	Circuit/layout weakness, timing margin, power/signal integrity
Test issue	Program/limit error, coverage gap, socket/contact or fixture
Foundry issue	Fabrication-related contact/via, metal, dielectric or process defect
Package issue	Bond/bump, RDL/substrate, solder or assembly-related defect
Handling issue	Evidence-linked ESD/EOS, mechanical, transport or use damage
Outstanding issue	Unresolved investigation: record evidence, hypotheses and next actions

Output: Mechanism, category, supporting evidence and uncertainty. Outstanding is a status; multiple causes may contribute. ESD/EOS evidence alone does not establish the event location.

10 Corrective action, validation and closure

Define containment and affected population. Assign actions to responsible teams. Validate effectiveness with relevant electrical/reliability checks; improve screening where justified. Close the FA/8D report with traceable evidence or explicit outstanding actions.

Output: Validated corrective action and closure, or documented ongoing investigation.